Unconstraint $\min f(x)$

Tylor Expansion $f(x) = f(x^*) + \langle \nabla f(x^*), d \rangle$

$+ \frac{1}{2} d^T H f(x^*) d + o(x^2)$ Direction $d = x - x^*$

1st NOC min $\Rightarrow \nabla f = 0$

For $x \in N_r(x^*)$, $f - f^* = \langle \nabla f(x^*), d \rangle + o(x) \geq 0$ for $\forall d$,

$\langle \nabla f(x^*), d \rangle$ must ≥ 0 for 1st order near enough, then ∇f^* must be 0

当 f 凸 x^* global $\Leftrightarrow \nabla f = 0$

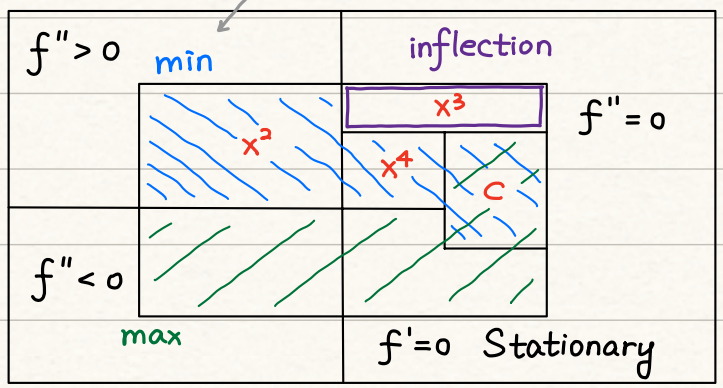
2nd NOC min $\Rightarrow H \geq 0$ 排除平台型

2nd SOC $H > 0$, $\nabla f = 0 \Rightarrow$ local isolated min

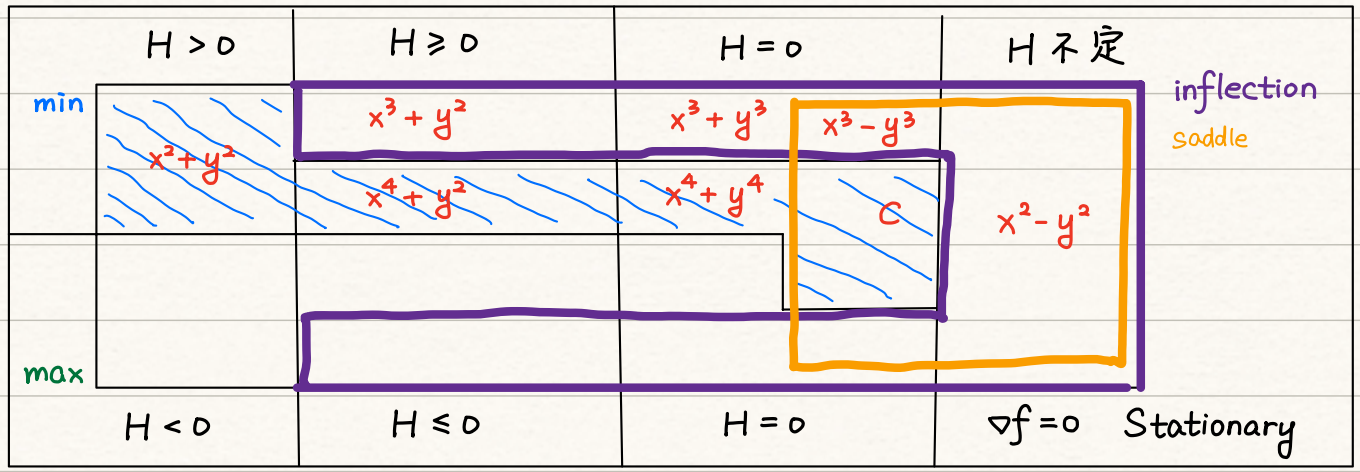
Inflection : $\nabla f = 0$, 非 min 非 max H 不定 $\Rightarrow$ Inflection

Single Variable

$f'' > 0$ 指在 x 点处，而 f convex 是 $\forall x$, $f'' \geq 0$



Multi Variable

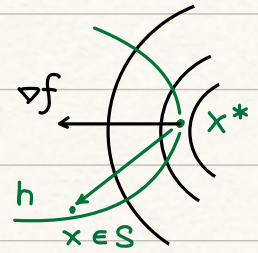


Constraint

$\min f(x) \quad \text{s.t. } h(x) = 0,\ \varphi(x) \le 0$

$\min f(x) \quad \text{s.t. } x \in S$

x^* global $\iff \langle \nabla f^*, x - x^* \rangle \ge 0$



Pf: $\Leftarrow$ f convex, $f(x) \ge f(x^*) + \langle \nabla f^*, x - x^* \rangle$

$\ge f(x^*)$ so x^* global min

$\Rightarrow$ Assume $\langle \nabla f^*, x - x^* \rangle < 0$

S convex, $\bar{x}_\alpha = \alpha x + (1-\alpha) x^* \in S$

$g(\alpha) = f(\bar{x}_\alpha)$, $g'(0) = \langle \nabla f^*, x - x^* \rangle < 0$

即沿 $x^* \to x$ 方向上, x^* 邻域上下降, 矛盾

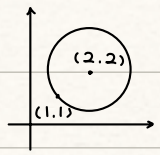
$L(x, \lambda, \mu) = f + \lambda h + \mu \varphi$

Regularity Point all ∇h, $\nabla \varphi$ independent (active) Standard $\dim h + \dim \varphi \le \dim x$ (active)

KKT 1st NOC min $\Rightarrow$

$\nabla L = 0$, $h(x) = 0$, $\mu \varphi(x) = 0$, $\mu \ge 0$, Regular

e.g. $\min f = (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{s.t. } x_1 + x_2 \le 2$



$L = f + \mu g = (x_1 - 2)^2 + (x_2 - 2)^2 + \mu (x_1 + x_2 - 2)$

$\nabla L = \begin{bmatrix} 2(x_1 - 2) + \mu \\ 2(x_2 - 2) + \mu \end{bmatrix} = 0 \qquad \mu (x_1 + x_2 - 2) = 0$

Case 1: $g = 0$, then $x_1 = 1$, $x_2 = 1$, $\mu = 2$

verify $\mu > 0$, feasible ✓

Case 2: $\mu = 0$, then $x_1 = 2$, $x_2 = 2$

verify $g(x) > 0$ infeasible

Regular verify $\nabla g(1,1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ 只有1个约束肯定线性独立

(P) convex problem, global min $\iff$ KKT

2nd NOC min $\Rightarrow d^T(HL)d \geq 0$, for $\forall$ feasible $d \neq 0$

such that $\langle \nabla h, d \rangle = 0$, $\langle \nabla g, d \rangle = 0$ (active)

2nd SOC KKT. $d^T(HL)d > 0$, for $\forall$ feasible d $\Rightarrow$ isolated min

Strong 2nd SOC KKT. $HL > 0$ $\Rightarrow$ isolated min

e.g. min $f = x_1^2 + x_2^2 - 3x_1x_2$ s.t. $x_1^2 + x_2^2 \leq 6$

$$HL = \begin{bmatrix} 2+2\mu & -3 \\ -3 & 2+2\mu \end{bmatrix} \quad \nabla g = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$$

3 KKT point : $(x_1, x_2, \lambda) = (0,0,0)$ $\underline{(\sqrt{3}, \sqrt{3}, \tfrac{1}{2})}$ $(-\sqrt{3}, -\sqrt{3}, \tfrac{1}{2})$ 举例

$$HL = \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix} \geq 0 \text{ 无结论}$$

确定 feasible d, $\langle \nabla g, d \rangle = 0$ so $d = \begin{bmatrix} d_1 \\ -d_1 \end{bmatrix}$

$$d^T(HL)d = d_1^2 (1, -1) \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 12 d_1^2 > 0, \ \forall d_1 \neq 0$$

由 2nd SOC, $(\sqrt{3}, \sqrt{3})$ 是 isolated min

e.g. min $f = (x_1 - 2)^2 + (x_2 - 2)^2$ s.t. $x_1 + x_2 \leq 2$

$$HL = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} > 0, \text{ isolated min}$$